Bala Sai Manikanta Sandeep Puppala

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Education

New York University — New York, NY
MS, Computer Science — 09/2026 – Expected 05/2028

Gandhi Institute of Technology and Management (GITAM) — Hyderabad, Telangana, India
B.Tech, Computer Science Engineering | CGPA: 3.79 — 08/2020 – 04/2024

Projects

Penalty Kick & Goalkeeper Direction Prediction from Video (PyTorch, YOLO, RTMPose, DINOv2) — 10/2026 – 11/2026

- Built an end-to-end video pipeline that predicts kick direction and goalkeeper dive from run-up footage ending **0.02 s before ball contact**, across **2,200 broadcast clips** from 215 source videos.
- Detected players and ball with **YOLO11x**, extracted 26-keypoint **RTMPose** skeletons for kicker and keeper, computed **optical flow** and geometry features, and automatically identified the kicker, keeper and camera view.
- Designed a **stacked ensemble** of 14 models: LightGBM on pose features plus fine-tuned **DINOv2 (ViT-B/L), ConvNeXt and R(2+1)D** video models, combined by logistic regression on out-of-fold predictions and validated with video-grouped 5-fold CV.
- Scored **0.826** on the public leaderboard vs. 0.514 baseline, with **92%** keeper accuracy; trained on HPC.

Cross-Modal Neuron Segmentation & Matching (PyTorch, Cellpose-SAM, DINOv2, OpenCV) — 09/2026 – 10/2026

- Segmented every neuron in paired **in-vivo two-photon** and **ex-vivo** fluorescence microscopy images of mouse brain, then matched each cell across the two modalities with no given alignment.
- Fine-tuned **Cellpose-SAM** and **DINOCell** (DINOv2 ViT-B/8 with a Cellpose flow head) for instance segmentation, using top-hat/CLAHE inputs, 3× upscaling, sliding-window flip TTA and random-convolution augmentation.
- Achieved a score of 0.531 on the public leaderboard, with a **0.548** leave-one-mouse-out held-out score; trained on HPC (A100, Slurm).

Experience

Storable India — Hyderabad, Telangana, India

Software Engineer I — 04/2025 – 07/2026

- Developed an AI-powered chatbot system to generate structured JSON outputs for dynamic section rendering, reducing section creation time from over **2 hours** to under a minute while implementing guardrails for response validation and workflow reliability.
- Built a Micro-Frontend architecture to integrate two web applications into the CMS with a configurable setup, enabling scalable deployment of an improved UI across **600+ facilities**.
- Received the quarterly **Storrior Spotlight Award** four times and the **Emerging Storrior Award** for exceptional performance, rapid growth, and consistent delivery of high-priority tasks.

Associate Software Engineer — 08/2024 – 03/2025

- Led the development of a compliance-related feature by consuming APIs from two products, enabling an on-demand sync every 3 hours and securely storing specific data in the database for seamless website access, preventing potential legal liabilities of **$10,000 per website** for 20% of sites.

Software Development Engineer - Intern — 02/2024 – 08/2024

- Enhanced the MERN-stack CMS supporting nearly **5,000 websites**, upgrading react-bootstrap and bootstrap to current major versions and resolving nearly 50 bugs in 2 weeks.
- Achieved 1st place in a Hackathon conducted by Storable for developing a Flutter application that consumes internal product data via APIs.

Skills

- **CV**: OpenCV, Object Detection, Pose Estimation, Optical Flow, Video Classification, Vision Transformers, ConvNeXt, R(2+1)D
- **ML & Deep Learning**: PyTorch, timm, ONNX Runtime, LightGBM, Scikit-learn, Ensembling & Stacking, Pandas, NumPy, Prompt Engineering, AI Agents
- **Compute & Cloud**: HPC, Mixed-Precision Training, AWS (S3, Bedrock), PostgreSQL, MongoDB, SQLite
- **Languages & Web**: Python, JavaScript, Java, C, Bash, React, Redux, GraphQL, Node.js, REST APIs, Flask
- **Tools**: Git, Linux, VS Code, Postman, VIM

Research

A Two-warehouse Inventory Model with an Advanced Payment Mechanism for Stock-Dependent Demand, Deteriorating Items with a Fixed Shelf Life and Partial Backlogging — 06/2024

- Developed a mathematical model for managing the inventory of perishable goods with real world complexities.

Achievements

- Ranked amongst Top **3%** candidates in Round I and Top **0.74%** candidates in Round II of IICC (Innovate India Coding Championship) (**1 Lakh** candidates participated) conducted by Chandigarh University, Coding Ninjas and AICTE.
- Secured 96.3469 percentile in JEE MAINS B.TECH.

Leadership & Activities

- **LEAD1x: Exercising Leadership: Foundational Principles** - Harvard University
- Finalist of Aspire Leaders Program Offered by Aspire Institute

Interests: Exploring new technologies by browsing online articles, enhancing my knowledge and staying current with industry trends.